

Workflow Reasoning (Metric A)

Workflow Reasoning evaluates whether the predicted diagnostic reasoning path matches the ground-truth workflow and whether the answers on each step are semantically correct.

Each reasoning step is a transition between a question and the next question in the chain. After canonicalization (normalization of capitalization, whitespace, and trailing punctuation), an edge is defined as:

(question, next_question)

Let the ground-truth edge set for a case be E_{gt} and the predicted edge set be E_{pred} .

Participants should use the same canonical question and next_question strings as in the training annotations wherever the metric requires them. Minor formatting differences may be normalized during evaluation; paraphrased workflow questions are not equivalent.

The Workflow Reasoning metric consists of four components.

1. Binary Path Validity (BPV)

Weight: 0.05

Binary Path Validity checks whether the predicted reasoning graph exactly matches the ground-truth reasoning graph.

$BPV = 1$, if $E_{pred} = E_{gt}$

$BPV = 0$, otherwise

where $1(.)$ is the indicator function.

2. Edge-F1

Weight: 0.30

Edge-F1 measures partial structural agreement between the predicted and ground-truth reasoning graphs.

$TP = |E_{pred} \cap E_{gt}|$

$FP = |E_{pred} \setminus E_{gt}|$

$FN = |E_{gt} \setminus E_{pred}|$

$Precision = TP / (TP + FP)$

$Recall = TP / (TP + FN)$

$Edge-F1 = (2 \times Precision \times Recall) / (Precision + Recall)$

This gives partial credit when part of the reasoning structure is correct, even if the full path is not an exact match.

3. MESS (Mean Edge Semantic Similarity)

Weight: 0.25 (non-final steps only)

MESS evaluates the semantic similarity between predicted and ground-truth answers on expected reasoning edges. The final pathology-report step is excluded from MESS.

For each ground-truth edge e :

if the same edge exists in the prediction, compute the cosine similarity between the ground-truth answer embedding and the predicted answer embedding

if the edge is missing, assign a score of 0

Formally:

$s(e) = \text{cosine}(\phi(a_{gt,e}), \phi(a_{pred,e}))$, if e is in E_{pred}

$s(e) = 0$, if e is not in E_{pred}

where:

$a_{gt,e}$ = ground-truth answer for edge e

$a_{pred,e}$ = predicted answer for edge e

$\phi(.)$ = biomedical text embedding function

$\text{cosine}(.)$ = cosine similarity

Then MESS is computed as:

$\text{MESS} = (1 / |E_{\text{nonfinal}}|) \times \sum s(e)$, for all e in E_{nonfinal}

where E_{nonfinal} is the set of ground-truth edges excluding the final pathology-report step.

In other words, MESS is the average semantic similarity over non-final ground-truth edges, with missing predicted edges receiving zero semantic credit.

4. Final Report Score (REG25-style)

Weight: 0.40

The final workflow step asks for the final pathology report. This submetric compares the predicted report answer on that step to the ground-truth report:

Final Report Score =

$0.15 \times (\text{ROUGE-L F1} + \text{BLEU-4 proxy})$

$+ 0.40 \times \text{keyword Jaccard}$

$+ 0.30 \times \text{embedding cosine}$

Keyword overlap uses biomedical entity keywords extracted from both texts. Embedding similarity uses mean-pooled PubMedBERT embeddings; values above 0.5 are rescaled into [0, 1]. The result is clamped to [0, 1].

5. Final Workflow Reasoning Score

For each case:

Workflow Reasoning Score =

$0.05 \times \text{BPV}$

$+ 0.30 \times \text{Edge-F1}$

$+ 0.25 \times \text{MESS}$

$+ 0.40 \times \text{Final Report Score}$

The leaderboard value is the unweighted mean of per-case Workflow Reasoning Scores over all ground-truth cases.

Short interpretation

Binary Path Validity checks exact path match

Edge-F1 checks partial structural overlap

MESS checks semantic correctness of intermediate reasoning answers

Final Report Score checks quality of the concluding pathology report

Workflow Reasoning Score combines all four into one final score